

Accepted Service Adaptation: Compact Allocation Quotients and Minimal Price Memory

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Abstract. A service provider can return to the same public operating state with different obligations inherited from past renewal reviews. We characterize when these obligations admit an exact finite-memory implementation. For positive additive payments and separable quadratic rewards without switching costs, the full-history problem is a classical laminar allocation problem. We prove an explicit occurrence-to-vertex quotient and show that normalized memoization yields a globally bounded family of payment responses on the public graph. A coefficient-event construction has polynomial rational bit complexity. A backward behavioral equivalence then characterizes the smallest writable alphabet implementing the unique optimum, including randomized exact-optimum controllers under an explicit information-accounting convention. For a heterogeneous renewal family, we derive an ordered memory-value frontier and prove that greater outside-option dispersion at a fixed mean payment strictly increases the loss from insufficient memory. A shared-table comparator has a graph-sized convex formulation, and fixed positive-release queries admit exact supporting cuts even at coincident bounds. Independent tree allocation, sparse convex optimization, and public-graph promise discretization provide algorithmic comparisons. Exact scale tests include more than two thousand public vertices and a separate breakpoint-dense family. The broader switching and certified-deployment theory is preserved in supplementary volumes.

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1 Introduction

A service agreement can return to the same public operating condition after different sequences of renewal reviews. Those histories need not permit the same future service. A customer who could have exited at an earlier review may have accepted a different remaining payment obligation. A provider that remembers only the current public condition can therefore lose value, even when demand is Markovian, prices are fixed, and neither party has private information. The operational question is not whether full history improves a particular schedule. It is how to construct an accepted optimal protocol, how much information its execution actually needs, and what an optimized memory restriction costs.

We answer these questions for positive additive payments, separable strictly concave quadratic operating rewards, and an exogenous recombining public graph. The principal algorithm excludes intertemporal switching costs and coupled vector commitments. These conditions make one accumulated payment price sufficient for optimal continuation. Each participation cap replaces prices below a threshold by that threshold. Execution retains the largest threshold encountered, together with a query-specific initial price. We construct the entire response family before choosing the initial price; the construction neither obtains an extensive-form optimum first nor enumerates a scenario tree.

The closest optimization connection is classical tree-constrained resource allocation. Mjelde’s tree constraints and Tang’s separable convex reductions are directly relevant, not merely analogies (Mjelde, 1983; Tang, 1990). After weighting each occurrence by its probability and discount factor, our full-history model belongs to that class. Moreover, memoizing the *normalized* subtree response produces the same graph recursion. We make this equivalence explicit rather than present marginal allocation or memoization as new principles. Our results quantify what the quotient representation makes available: a graph-sized universe of response breakpoints, a polynomial rational-arithmetic implementation, and an exactly minimal execution alphabet for a fixed contractual query. These are separate statements about representation, computation, and implementation.

The paper develops three complementary results. First, an occurrence-to-vertex theorem identifies the precise information discarded by the quotient. A coefficient-event implementation then gives explicit arithmetic, storage, and rational bit bounds. Second, a backward equivalence relation merges price states exactly when they prescribe the same complete future actions. This relation characterizes the smallest writable alphabet for an optimal protocol and shows that its classes are ordered by incoming price. Strict concavity extends the exact-optimum lower bound to randomized controllers under the same accounting of retained information. Third, a renewal family admits a complete value–memory frontier with heterogeneous probabilities and operating costs. At a fixed mean payment, increasing outside-option dispersion strictly raises the loss from an insufficient memory alphabet, although arbitrarily small dispersion can already change the exact alphabet requirement.

An optimized public table supplies the economic comparator. At zero release it has a graph-sized convex formulation. At a fixed positive release, the price compiler supplies an exact value and a supporting inequality, including coincident physical and contractual bounds. We do not equate an exact oracle with finite termination of the continuous outer table-design problem.

The computational study separates implementation verification from comparison with other algorithms.

It includes an independently implemented occurrence-indexed laminar reduction, a generic sparse convex optimizer on unfoldable instances, and a primal promise-grid method that already works on the public graph. Larger examples distinguish response-table growth from rational coefficient growth. No claim of superiority over all policy-graph or stochastic dual dynamic programming methods is inferred from the size of a hypothetical scenario tree. The broader accepted-adaptation results on switching, comparator-adjusted rents, continuous states, and certified deployment remain available in the retained theory volumes, with their statements and proofs unchanged; they are not used to enlarge the domain of the finite-price theorem.

2 Service Agreements and Their Laminar Allocation Form

2.1 Public primitives and accepted continuations

Let $G = (\mathcal{V}, \mathcal{E})$ be a finite, rooted, acyclic public graph. Vertices include date; all are reachable. Write $N = |\mathcal{V}|$, $M = |\mathcal{E}|$, and let $p_{vw} > 0$ be the exogenous probability on edge (v, w) , with probabilities summing to one at a nonterminal vertex. The discount factor is $0 < \beta \leq 1$. An occurrence h is a root-to-vertex history in the finite unfolding $\mathcal{T}$; its public vertex is $v(h)$ and its discounted reach weight is $w_h = \beta^{t(h)} \Pr(h) > 0$. Write $j \succeq h$ for a descendant, including h , and $H = |\mathcal{T}|$.

At occurrence h of v , the provider chooses a tier $x_h \in [l_v, h_v] \subseteq [0, 1]$. The payment coefficient $a_v > 0$ is fixed, and the operating reward is $g_v(x) = r_v x - q_v x^2/2$, with $q_v > 0$. Customer utility is a publicly specified service-benefit term minus $a_v x$. For example, fixed stock s_z yields the stage utility

$$u(z, s_z, x) = v \mathbb{E} \min(s_z, D_z) - a_z x, \quad a_z = r_z^{\text{premium}} - \kappa \mathbb{E}(D_z - s_z)^+ > 0. \quad (1)$$

Stock may be stipulated, or identified by the service and cost commitments in the retained stock-identification results. The customer can leave before each modeled review; the provider commits to its contingent protocol. Retaining the customer requires the conditional service benefit minus remaining payment to exceed the public outside continuation. Assume their difference is a vertex-sufficient cap B_v . The accepted optimization is

$$\begin{aligned} J(b_0) = \max_x \quad & \sum_{h \in \mathcal{T}} w_h g_{v(h)}(x_h) \\ \text{subject to} \quad & Y_h := \sum_{j \succeq h} \frac{w_j}{w_h} a_{v(j)} x_j \leq B_{v(h)} \quad (h \in \mathcal{T}), \quad Y_0 = b_0, \\ & l_{v(h)} \leq x_h \leq h_{v(h)}. \end{aligned} \quad (2)$$

The root has a cap as well as the equality promise. No privately informed reports, endogenous termination choice, or policy-dependent transition probabilities are part of this institution. The no-switching restriction is essential to the scalar-price result; the retained general model continues to allow switching and is not replaced by (2).

The feasible remaining-payment interval is computed without unfolding:

$$L_v = a_v l_v + \beta \sum_w p_{vw} L_w, \quad U_v = \min \left\{ B_v, a_v h_v + \beta \sum_w p_{vw} U_w \right\}. \quad (3)$$

A subtree is feasible precisely when its successors are feasible and $L_v \leq U_v$. Throughout, $b_0 \in [L_0, U_0] \subseteq (-\infty, B_0]$. Convexity fills every point between the two extreme payments. Compactness gives an optimum and strict concavity gives a unique occurrence-indexed action vector.

2.2 An exact mapping, not an analogy

Proposition 2.1 (Laminar allocation equivalence). *Define $z_h = w_h a_{v(h)} x_h$. Problem (2) is exactly the separable concave allocation problem*

$$\begin{aligned} \max_z \quad & \sum_h \left[\frac{r_{v(h)}}{a_{v(h)}} z_h - \frac{q_{v(h)}}{2w_h a_{v(h)}^2} z_h^2 \right], \\ & \sum_{j \succeq h} z_j \leq w_h B_{v(h)}, \quad \sum_j z_j = b_0, \\ & w_h a_{v(h)} l_{v(h)} \leq z_h \leq w_h a_{v(h)} h_{v(h)}. \end{aligned} \quad (4)$$

The constrained descendant sets are laminar. The change of variables is invertible and preserves objective values and optimal policies.

Proof. Substitution gives each term and row of (4). Two rooted descendant sets in a tree are either disjoint or one contains the other. Positivity of $w_h a_{v(h)}$ makes the coordinate map bijective and preserves every inequality direction. $\square$

Thus the expanded primal lies in the tree-constrained allocation lineage of Mjelde (1983) and Tang (1990). Tang reports at most twice the number of variables in one-variable convex subproblems. That count is not a count of elementary rational operations for a complete parametric response family. The chain-nested allocation results of Vidal et al. (2016); Schoot Uiterkamp et al. (2021); Wu et al. (2021) concern related but different input structures. In particular, a complexity bound for an ascending chain cannot simply be assigned to arbitrary branching descendant sets.

After subtracting lower allocations, the packing region is a laminar polymatroid, and the exact total selects a base of its truncation (Electronic Companion, Section EC.1). This connects the model to the broader allocation literature, including Federgruen and Groenevelt (1986) on discrete greedy allocation and Groenevelt (1991) on separable concave maximization over polymatroids. These connections identify inherited feasible-set geometry; they do not by themselves provide a compact-DAG response-size or writable-memory bound. Nor do the discrete polynomial bounds stated in that literature's abstracts establish the binary-input bound proved here for the continuous quadratic response construction.

What memoization does and does not change. An occurrence-indexed marginal reduction may represent its subtree payment in the unnormalized units z . Its entire response is the occurrence weight times a common conditional response, proved below. Dividing by that weight makes occurrences of the same public vertex identical. Memoizing those normalized responses therefore gives our graph recursion. There is no additional stochastic obstruction or different first-order condition. The quantitative questions are whether complete responses have a bounded common knot universe, whether exact coefficients have polynomial bit length, and how the resulting policy can be minimized as an execution machine. Theorems 3.1–3.3 and 4.1 answer those questions directly. We do not assert that scalar marginal allocation, subtree reduction, or the principle of memoization originated here.

An explicit-tree implementation has input size H and must at least read its H local records; a compact representation has N vertices and M edges. This is an input/output comparison, not a lower bound against every algorithm that accepts a compact graph. The independent tree baseline below implements a specified exact reduction; it is not presented as the original authors’ implementation or as the fastest available laminar algorithm.

2.3 Relation to contractual states and policy graphs

Promised continuation utility is a standard recursive state in dynamic contracting (Spear and Srivastava, 1987; Thomas and Worrall, 1988; Chen et al., 2020). Our remaining payment is a primal contractual state; an incoming payment price is its dual counterpart. The public-information institution does not solve the incentive or adverse-event allocation problems in Chen et al. (2020); Liang et al. (2023). It instead determines what accepted service adaptation requires after public continuation constraints have been imposed.

Policy graphs and stochastic decomposition already avoid naive scenario enumeration (Pereira and Pinto, 1991; Girardeau et al., 2015; Dowson, 2020). We compare an explicit primal promise-grid policy graph below, rather than treat the mere use of a public graph as a contribution. Our exact closure depends on one scalar additive commitment and separable quadratics; general convex policy-graph decomposition has a broader domain. The supporting cuts for fixed tables are also instances of standard dual decomposition (Benders, 1962; Boyd and Vandenberghe, 2004), not a new general finite-convergence theorem.

3 A Quantitative Quotient of the History Tree

3.1 Conditional responses and cap reflection

For an occurrence h of v , let $H_h(\eta)$ be the maximum conditional reward minus ηY_h , imposing all subtree caps, but not an exact payment equality at h . Let $H_h^{\text{pre}}(\eta)$ remove only the cap at h , retaining every successor cap. Conditional rewards use weights w_j/w_h . Define $R_h(\eta)$ as the payment of the unique maximizer of $H_h(\eta)$; use $d_h(\eta)$ for the pre-cap payment. These definitions separate a removed current cap from the constrained continuation problems below it.

Theorem 3.1 (Occurrence-to-vertex quotient and finite response closure). *Under the assumptions of Section 2, occurrences h, h' of a public vertex v with the same incoming price η have identical normalized*

optimal continuation plans. In particular, $H_h = H_{h'} =: H_v$, $H_h^{\text{pre}} = H_{h'}^{\text{pre}} =: H_v^{\text{pre}}$, and $R_h = R_{h'} =: R_v$. The following backward construction computes the common payment responses and optimal policies:

$$x_v^0(\eta) = [(r_v - a_v\eta)/q_v]_{[l_v, h_v]}, \quad (5)$$

$$d_v(\eta) = a_v x_v^0(\eta) + \beta \sum_w p_{vw} R_w(\eta), \quad (6)$$

$$R_v(\eta) = \min\{B_v, d_v(\eta)\}. \quad (7)$$

Write $d_v^- = a_v h_v + \beta \sum_w p_{vw} U_w$ for the finite constant on the left tail of d_v . If $B_v \geq d_v^-$, assign no barrier at v . Otherwise let α_v be the leftmost finite solution of $d_v(\alpha_v) = B_v$. The optimal local action and the incoming price sent to every successor are determined by

$$s_v = \max\{\eta, \alpha_v\}, \quad x_v = x_v^0(s_v), \quad (8)$$

where a missing barrier leaves $s_v = \eta$.

Every R_v is continuous, nonincreasing, piecewise affine with constant tails and range $[L_v, U_v]$. The global universe of distinct response knots has size at most $3N$. Complete response tables can be constructed in

$$A = O((N + M)N \log(N + 1)) \quad (9)$$

arithmetic operations and comparisons, using $O(N^2 + M)$ scalar storage. For a fixed root promise, choose any finite η_0 with $R_0(\eta_0) = b_0$. Execution uses at most $N + 1$ distinct incoming-price labels, drawn from η_0 and the finite cap barriers. The construction does not enumerate H occurrences.

Proof. The subtree below an occurrence is determined by its displayed vertex and subsequent public transitions. The map replacing the initial occurrence by another occurrence of the same vertex preserves all suffix paths, relative probabilities, discounts, local rewards, bounds, and caps. It is consequently a bijection of the normalized feasible problems at any common η . Strict concavity makes their maximizing action vectors agree. This proves the quotient independently of the algorithm.

Remove the current cap. Separability gives (5) and independent successor problems at the same incoming price, proving (6). Their unique maximizers have continuous payment responses. Backward induction starts with a clipped affine local payment and preserves continuity, monotonicity, and affine pieces under weighted summation. Its two tails are the extreme attainable payments before the current cap.

When the current cap is redundant, this pre-cap maximizer is feasible. Otherwise feasibility implies that B_v belongs to the range of d_v . The clipped finite response reaches its tails at finite prices, so a finite leftmost crossing exists. For $\eta < \alpha_v$, the pre-cap optimum at α_v has payment B_v . Let z^* be that optimum and z any policy satisfying the current cap. Writing G for conditional reward and Y for payment gives

$$G(z) - \eta Y(z) \leq G(z^*) - \alpha_v B_v + (\alpha_v - \eta) Y(z) \leq G(z^*) - \eta B_v.$$

For $\eta \geq \alpha_v$, the pre-cap optimizer already satisfies the cap. These two cases prove reflection, (7), and its implementing policy. If the payment is constant between two prices, the two optimality inequalities imply

that either unique optimizer is also optimal at the other price. Thus the policy is unchanged on an equality plateau even when the supporting dual price is not unique. This also covers a cap at the minimum feasible payment and fixed local intervals.

Local clipping contributes at most two knots. Summing child responses introduces no knot outside their union. Capping can remove knots and introduces at most one crossing. Inducting over distinct *vertices*, rather than occurrences, gives a global union of at most $3N$. Each of the N responses can still have order N segments, so total response storage is quadratic, not linear. Directly merging weighted coefficient changes at sorted knots gives (9); the graph takes an additional $O(N + M)$ records.

Finally, each executed price is the maximum of the root price and the barriers along the realized path. There are at most $N + 1$ such numerical labels for this fixed root query, and at most $N(N + 1)$ reachable vertex–price pairs. The root maximizer at η_0 has precisely b_0 , so it also maximizes reward among policies satisfying the equality. $\square$

The unnormalized response at occurrence h in Proposition 2.1 is $w_h R_{v(h)}(\eta)$. This identity is the explicit link to an occurrence-indexed tree reduction. A normalized memoized implementation enjoys the same quotient bound; it is not an algorithmic competitor that our implementation is claimed to dominate.

Corollary 3.2 (Several coordinates under one scalar commitment). *With d_v separable local coordinates, positive payment coefficients, positive quadratic coefficients, and a product interval, let $D = \sum_v d_v$. The knot universe has size at most $2D + N$, response storage is $O(N(2D + N) + M)$, and the incoming-price alphabet still has at most $N + 1$ elements for a fixed query.*

Proof. There are two clipping knots per coordinate, but only one scalar cap barrier per vertex. Clipping knots describe how a local action responds to a given price; they are not additional prices written into the running record. The common commitment price and cap reflection are unchanged. $\square$

Independent commitments can be separated only when rewards and feasible sets also separate. Coupled vector promises, nonseparable quadratic terms, and intertemporal switching do not satisfy the scalar reflection argument. Their retained primal promise-state and convex formulations remain applicable.

3.2 Rational bit complexity

An arithmetic count alone does not establish polynomial time on binary inputs. Let L be the total binary encoding length of all rational primitives, graph indices, and the rational root query, with rationals in lowest terms. Bit bounds below concern the coefficient-event construction, which sums affine coefficients directly. A newly computed cap crossing is used to locate an interval, *not* as an input to coefficient interpolation.

Theorem 3.3 (Polynomial rational preprocessing). *All response slopes, intercepts, finite knots, cap barriers, and a returned finite root price can be represented with*

$$S = O(NL + \log(N + M + 1)) \quad (10)$$

bits per numerator and denominator. The construction in Theorem 3.1, followed by one root inversion, admits the conservative bit-operation bound $O(AS^3)$ and output storage $O((N^2 + M)S)$ bits. These are polynomial bounds for rational inputs, not a strongly polynomial claim or a claim that coefficient length is uniformly bounded in the horizon.

Proof. Fix an open interval between global knots. At a vertex whose current cap binds on that interval, the payment response is the constant B_v . Otherwise its coefficients are a local affine payment plus the discounted, probability-weighted coefficients at successors. The possible nonzero local sources are

$$B_v, \quad a_v l_v, \quad a_v h_v, \quad a_v r_v / q_v, \quad -a_v^2 / q_v. \quad (11)$$

Expanding either coefficient through the acyclic graph therefore gives a sum over directed paths of one source multiplied by edge probabilities and a power of β . Expansion stops at a capped vertex. No path repeats an edge, and its length is at most $N - 1$.

Let D_0 be the product of all positive denominators and the absolute values of all nonzero numerators appearing in the rational primitive input. Its logarithm is $O(L)$. Every path-term denominator divides D_0^{N+4} : this exponent covers repeated discounting, each edge probability, and the products and divisions in (11). Hence a common denominator has $O(NL)$ bits even though the number of paths may be exponential. The sum of nonnegative discounted path weights to source occurrences is at most N , because it is bounded by the expected number of visited vertices. The absolute value of a source is at most $2^{O(L)}$. Thus the magnitude of a coefficient is at most $N2^{O(L)}$; multiplying by the common denominator bounds its numerator by (10).

Direct merging uses sums and differences of such adjacent coefficients. Its intermediate partial sums add at most a polynomial factor in $N + M$ to the preceding magnitude bound and have the same kind of path-product denominator. A new cap crossing has form $(B_v - c)/m$ on a nonconstant line $m\eta + c$; local clipping knots and a root inverse are ratios of a constant number of the same bounded-height rationals. Consequently their heights also satisfy (10). Exact comparisons use cross products of bounded-height integers. Elementary integer arithmetic and reduction by greatest common divisors cost at most $O(S^3)$ per rational operation under a conservative implementation. Multiplying by (9) proves the stated time bound; counting stored scalars proves the storage bound. $\square$

The theorem bounds preprocessing and inversion, not the decimal length of a complete printed primal–dual certificate by the same S . Different realized barriers can introduce different denominators into a sum of rewards. A straightforward polynomial bound for aggregate execution values is larger; the companion gives one and the experiments separately time certificate generation and audit. This distinction is practically relevant even when each response coefficient is modest.

3.3 A compact exact certificate

For each reached (v, η) , propagate discounted occupation weight w , effective price s , action x , and multipliers $\chi = s - \eta$, $\ell, u \geq 0$. They satisfy

$$r_v - q_v x - a_v s + \ell - u = 0, \quad \ell(x - l_v) = u(h_v - x) = \chi(B_v - R_v(\eta)) = 0. \quad (12)$$

Weight $\beta w p_{vw}$ moves to (w, s) . A backward conditional-payment calculation verifies every cap; forward flow verifies occupation weights and the root promise. The resulting dual upper bound is

$$\eta_0 b_0 + \sum_{(v, \eta)} w \left[\chi B_v + u h_v - \ell l_v + \frac{(r_v - a_v s + \ell - u)^2}{2q_v} \right]. \quad (13)$$

Ancestor prices telescope under the flow equations, and (12) makes (13) equal the implemented reward. These rational checks certify a particular optimum; verifying entire response-curve identities is a separate implementation test. Neither kind of code execution replaces the mathematical proof above.

4 Minimal Execution Memory and Its Contractual Value

4.1 What the controller may remember

Fix the graph, all primitives, and one root promise. At a decision, the decoder observes the current public vertex (including date) and a writable symbol. A read-only table may contain primitives, thresholds, the query's root price, and the transition/output maps. Between decisions the update map may depend on the old symbol and the realized public transition; the next action decoder has no separate access to the past transition. A retained promise, past tier, branch identity, or random seed must be charged as writable information if it carries history not in the current public vertex. In particular, a continuous remaining promise is not an uncharged alternative memory register. This is a specified controller architecture, not a claim about every possible hardware observation interface.

The price construction supplies at most $N + 1$ symbols. This counts a writable *index*, requiring at most $\lceil \log_2(N + 1) \rceil$ bits, not the read-only graph, response tables, transition maps, or numerical precision. Changing the root promise requires a new inversion and may change the set of reachable states. If an inherited physical tier is separately observable without charge, the equivalence relation below must be conditioned on that enlarged public observation; the minimality statement is then a different one.

Let $\mathcal{P}_v$ be the incoming prices reachable at v under Theorem 3.1. For $\eta \in \mathcal{P}_v$, put $s_v(\eta) = \max\{\eta, \alpha_v\}$ with the missing-barrier convention. Define an equivalence relation backward:

$$\eta \sim_v \eta' \iff \begin{cases} x_v^0(s_v(\eta)) = x_v^0(s_v(\eta')), \\ s_v(\eta) \sim_w s_v(\eta') \text{ for every } (v, w) \in \mathcal{E}. \end{cases} \quad (14)$$

At a terminal vertex only the first condition is needed. Child prices in this display are reachable because every displayed edge has positive probability.

Theorem 4.1 (The minimal price-state execution alphabet). *Let c_v be the number of classes of $\sim_v$ on $\mathcal{P}_v$. Under the preceding observation model, the minimum size of a common writable alphabet implementing the unique optimal policy is*

$$K_* = \max_{v \in \mathcal{V}} c_v. \quad (15)$$

Each class is contiguous in the ordered list of incoming prices at its public vertex. A backward signature

construction computes the classes on the reached vertex–price graph in $O((N + M)N \log(N + 1))$ arithmetic comparisons and elementary index operations, with $O((N + M)N)$ explicit transition/signature storage. The exact-optimum lower bound also holds for randomized controllers whose retained private randomness is included in the same memory accounting and whose feasibility constraints hold in conditional expectation at each public history.

Proof. Induction on remaining horizon shows that (14) holds exactly when two incoming prices induce the same action at every future suffix history. One class can therefore use a single label: its output is the common action and its update on edge (v, w) is the corresponding child class. These maps are well defined by (14). Class indices can be reused across different public vertices because the decoder observes that vertex, proving sufficiency of (15).

If two occurrences of the same public vertex require different future action plans, assigning them the same writable symbol would give the same output and update rules from that point onward. Some positive-probability suffix would then receive the wrong action. Strict concavity makes the globally optimal occurrence action vector unique; hence such a controller cannot attain $J(b_0)$. This proves necessity.

Along a fixed suffix path, the effective price is the maximum of the incoming price and a fixed finite set of barriers. Every local action is nonincreasing in that effective price. Thus every coordinate of the complete future plan is nonincreasing in the initial incoming price. If two ordered endpoint prices have equal plans, every intervening price has the same plan coordinate by coordinate. This proves contiguity. A signature consists of the local action and the tuple of successor-class indices. Sort or compare signatures in reverse topological order. There are at most $N(N + 1)$ reached pairs and at most $M(N + 1)$ pair transitions, yielding the stated bound without unfolding suffixes.

For a randomized feasible controller, take the mean action at each public history. Exogenous transitions and linear payments/caps make this mean vector feasible. Strict concavity gives

$$\sum_h w_h \mathbb{E} g_{v(h)}(X_h) \leq \sum_h w_h g_{v(h)}(\mathbb{E} X_h) \leq J(b_0),$$

with equality only if each action is almost surely constant at the unique optimal value. Distinct required continuation plans must consequently have disjoint supports of retained symbols; a shared symbol with positive probability would impose the same future behavior on both histories. Additional retained randomness that could distinguish them is memory under the stipulated model. Therefore at least c_v symbols are needed at v even with randomization. $\square$

The theorem minimizes exact *execution* memory for this fixed optimum; it does not minimize the response compiler’s storage or characterize the best approximate policy for every alphabet size. Those are different design questions. Randomization is covered for attaining the exact optimum, not asserted to leave every restricted-memory value unchanged.

4.2 Ordered memory frontiers beyond equal branch probabilities

Consider three dates. The root tier is fixed at zero. At the intermediate date a branch j is observed with probability $\pi_j > 0$, where $\sum_j \pi_j = 1$. All branches then enter the same terminal public vertex. Payments and discounting equal one. Intermediate and terminal tiers are in $[0, 1]$. Branch j has cap $0 < b_1 < \dots < b_k < 1$, and the root promise is $\bar{b} = \sum_j \pi_j b_j$. At the intermediate date the reward is $-h_j(y)$, where h_j is convex, nondecreasing, and $h_j(0) = 0$. The common terminal reward $f(c)$ is continuous, concave, and strictly increasing on $[0, 1]$. The terminal cap is one. The terminal decoder observes only its public vertex and its memory symbol, not the inherited intermediate tier.

Theorem 4.2 (Heterogeneous renewal frontier). *The unique full-information policy is $y_j = 0$, $c_j = b_j$, with value $J = \sum_j \pi_j f(b_j)$. Exact implementation requires k symbols. With at most m deterministic terminal symbols, an optimal encoding has contiguous nonempty cells in cap order. For a cell $i, \dots, j$, its terminal tier is b_i and its loss is*

$$C(i, j) = \sum_{h=i}^j \pi_h [f(b_h) - f(b_i) + h_h(b_h - b_i)]. \quad (16)$$

The exact minimum loss is $D_m = D(\min(m, k), k)$, where

$$D(s, j) = \min_{s-1 \leq i < j} \{D(s-1, i) + C(i+1, j)\}, \quad D(0, 0) = 0, \quad (17)$$

and all other infeasible entries equal $+\infty$. With a precomputed cell-cost table this requires $O(mk^2)$ arithmetic operations. For quadratic f and quadratic h_j , prefix sums of the probability-weighted coefficients compute each cell cost in constant arithmetic time.

Proof. The root promise is the mean of the branch caps, so every branch cap must bind. Hence $y_j = b_j - c_j$ and $0 \leq c_j \leq b_j$. The branch reward $f(c) - h_j(b_j - c)$ is strictly increasing on that interval. Full information therefore uniquely selects $c_j = b_j$; different terminal actions require different symbols.

Within a memory cell all terminal actions coincide and cannot exceed its smallest cap. Monotonicity selects that cap and subtraction from the full-information reward gives (16). Fix any collection of cell minima. Every branch strictly prefers the largest eligible selected minimum, irrespective of its probability and intermediate cost. Reassigning branches in this way keeps each selected minimum in its own nonempty cell and produces contiguous cells without reducing value. Splitting at the final cell gives (17). Expanding the quadratic terms gives the stated prefix-sum implementation. $\square$

This strengthens, and includes, the quadratic equal-probability frontier. With $f(c) = 2c - c^2/2$, $h_j(y) = y^2/2$, and $\pi_j = 1/k$, (16) becomes

$$C(i, j) = \frac{2 - b_i}{k} \sum_{h=i}^j (b_h - b_i). \quad (18)$$

For $k = 8$ and $b_j = j/9$, one, two, and three symbols lose $119/162$, $5/18$, and $49/324$, respectively. Two writable bits permit four symbols, not three. The segmentation recurrence is standard; the economic loss

and the ordered-cell reduction are the substantive statements.

The earlier nonsaturated two-branch example also remains useful. With branch caps $1/4$ and 1 , root promise $1/2$, and the same quadratic rewards, the full optimum uses terminal tiers $1/4$ and $3/4$ and has value $27/32$. An optimized public-only table uses common terminal tier $1/4$ and intermediate tiers $0, 1/2$, giving $13/32$. The adaptation gain is $7/16$ relative to the *optimized* restriction. At the full optimum both inherited intermediate tiers are zero, so making that tier observable does not evade this example's exact-memory lower bound. By contrast, the entire restricted-memory frontier above depends on not allowing a branch-specific intermediate tier to serve as an uncharged code.

4.3 Outside-option dispersion at a fixed mean payment

The next result connects a contractual primitive to memory value. Fix positive probabilities, an interior mean $\bar{b}$, and strictly increasing numbers d_j with $\sum_j \pi_j d_j = 0$. Let

$$b_j(\varepsilon) = \bar{b} + \varepsilon d_j, \quad 0 < b_j(\varepsilon) < 1. \quad (19)$$

Keep the common service-benefit continuation fixed. The outside options are that continuation minus $b_j(\varepsilon)$; increasing ε scales their dispersion while preserving the mean payment and mean outside option. Physical public transitions are unchanged.

Proposition 4.3 (Dispersion raises the value of contractual memory). *For the quadratic renewal family in (18), with arbitrary positive branch probabilities, every insufficient deterministic alphabet $m < k$ has $D_m(\varepsilon) > 0$, strictly increasing in $\varepsilon > 0$ over the admissible range. Moreover $D_m(\varepsilon) = O(\varepsilon)$ as $\varepsilon \downarrow 0$. Exact execution needs one symbol at $\varepsilon = 0$ and k symbols at every $\varepsilon > 0$.*

Proof. For a fixed contiguous partition with cell minimum i , the loss on branch h in that cell is

$$\pi_h \varepsilon (d_h - d_i) (2 - \bar{b} - \varepsilon d_i).$$

For $h > i$, its derivative is $\pi_h (d_h - d_i) (2 - \bar{b} - 2\varepsilon d_i) > 0$. If $d_i \leq 0$ this is immediate. If $d_i > 0$, admissibility implies $\varepsilon d_i < 1 - \bar{b}$, so the last factor exceeds $\bar{b} > 0$. A partition with fewer than k cells has at least one such positive term. Every one of the finitely many candidate partition costs is therefore strictly increasing and positive. Their minimum is strictly increasing as well: use the minimizing partition at the larger parameter to compare the two parameter values. Each cost is $O(\varepsilon)$ near zero. The exact alphabet statement follows from distinct optimal terminal tiers, with all tiers identical at zero dispersion. $\square$

An exact memory requirement can thus jump while its economic value tends continuously to zero. A retention technology should be evaluated against D_m , not solely against the number of distinguishable optimal tiers. This implication depends jointly on intermediate renewal rights, recombination at a common public state, and a fixed aggregate commitment; it is not obtained from the physical Markov state alone.

5 The Optimized Shared-Table Comparator

A prescribed observation restriction is optimized before its adaptation value is assessed. Let $c(v) \in \{1, \dots, C_{\text{info}}\}$ be its information cell and let u_c be a single table chosen before uncertainty. The restricted value is denoted K^{tab} , keeping the number of cells distinct from the value.

Proposition 5.1 (Zero-release graph formulation). *The optimized zero-release comparator is the convex quadratic program*

$$\begin{aligned} K^{\text{tab}} = \max_{u, y} \quad & \sum_v \bar{w}_v g_v(u_{c(v)}) \\ & y_v = a_v u_{c(v)} + \beta \sum_w p_{vw} y_w, \quad y_v \leq B_v, \quad y_0 = b_0, \\ & l_v \leq u_{c(v)} \leq h_v. \end{aligned} \quad (20)$$

Here $\bar{w}_v = \beta^{t(v)} \Pr(v)$ is the graph's forward reach weight. There are $C_{\text{info}} + N$ variables and $O(N + M + C_{\text{info}})$ coefficient occurrences. For a fixed shared table, an incoming absolute switching cost can also be represented exactly by its root term and one reach-weighted epigraph term per public edge.

Proof. A common table fixes every local action by the public vertex. Backward conditional expectation gives the displayed y_v at every occurrence of v . Thus all history caps are equivalent to the N displayed caps. Conversely, every feasible table and payment vector implements accepted actions at all histories. Summing rewards by public reach weights proves the objective identity. An edge uses two fixed table entries, so its switching cost is also determined without additional history. $\square$

This supporting formulation uses standard convex optimization; it is not the central algorithmic novelty. In particular, a public table is not the full-information policy class. The difference $J(b_0) - K^{\text{tab}}$ compares two optimized protocols on identical primitives.

At a fixed table and release $\delta \geq 0$, permit $|x_h - u_{c(v(h))}| \leq \delta \omega_{c(v(h))}$, where $\omega_c \geq 0$. Set

$$l_v^{\text{eff}} = \max(l_v, u_{c(v)} - \delta \omega_{c(v)}), \quad h_v^{\text{eff}} = \min(h_v, u_{c(v)} + \delta \omega_{c(v)}).$$

Empty intervals or an infeasible root promise are rejected. Otherwise the finite-price compiler solves this fixed query exactly.

Proposition 5.2 (Supporting cuts including coincident bounds). *At every feasible fixed query (u, δ) , physical and corridor multipliers can be split so that nonnegative corridor multipliers $\rho_{v,\eta}^-, \rho_{v,\eta}^+$ give*

$$G_c = \sum_{(v,\eta): c(v)=c} w_{v,\eta} (\rho_{v,\eta}^+ - \rho_{v,\eta}^-), \quad H_\delta = \sum_{(v,\eta)} w_{v,\eta} \omega_{c(v)} (\rho_{v,\eta}^+ + \rho_{v,\eta}^-). \quad (21)$$

For any other feasible design $(\tilde{u}, \tilde{\delta})$,

$$V(\tilde{u}, \tilde{\delta}) \leq V(u, \delta) + G^\top (\tilde{u} - u) + H_\delta (\tilde{\delta} - \delta). \quad (22)$$

This statement includes physical/corridor ties and zero-width intersections.

Proof. At a reached pair let $d = r_v - q_v x - a_v s$. Choose effective lower and upper multipliers $\ell = \max(-d, 0)$, $u^+ = \max(d, 0)$. A positive lower multiplier occurs only when $x = l_v^{\text{eff}}$; a positive upper multiplier only when $x = h_v^{\text{eff}}$. Among the original lower inequalities, at least one attains l_v^{eff} . Assign nonnegative portions summing to ℓ only to the attaining physical and corridor inequalities. Do the analogous split for u^+ . At a tie any such split works. At zero width both lower and upper normal directions are available, and the sign choice still satisfies stationarity; no division by the width is used.

All assigned original multipliers satisfy complementarity because each receiving bound is active. They therefore form full fixed-query Lagrange multipliers with the root price and reflected cap multipliers. Keep them fixed while changing only the corridor right-hand sides. A corridor upper bound contributes $\rho^+(u_c + \delta\omega_c)$ and a lower bound contributes $-\rho^-(u_c - \delta\omega_c)$ to the dual objective. Their change gives (21). Weak duality and exact equality at the anchor prove (22).

The certificate’s old price labels merely aggregate the original history multipliers. Those multipliers can be assigned to the same public histories under any candidate policy, because the public process is exogenous. Thus the upper bound applies to every new feasible policy, not just policies sharing the old state labels. $\square$

The oracle supplies a supergradient of the fixed-table value in the design parameters. A continuous Benders-style outer master can use these cuts and feasible-query lower bounds, but no finite exact cut count or global positive-release polynomial algorithm follows. The exact graph-sized zero-release formulation, exact fixed-query compiler, and absence of a finite-cut guarantee for the outer design are distinct results.

6 Computational Evidence

6.1 Comparisons, measurement, and audit scopes

All instances are synthetic rational service agreements. The binary comparison family uses discount and payment coefficients one, two public regimes per noninitial date, independent seeded quadratic rewards, and continuation caps above a known feasible constant-tier policy. Thus an exact root promise is available without using any optimizer. The horizon ranges from four to ten dates for the unfolded comparisons, with two seeds at each horizon. A separate grid comparison extends to twelve dates. Every input and implemented policy is retained.

We implement four explicitly defined methods. The quotient compiler merges coefficient events on the public graph. The independent tree method unfolds the same input, changes variables as in Proposition 2.1, and performs occurrence-indexed exact marginal reduction without public-vertex memoization. It uses a separate interpolation implementation, not the quotient code. The generic numerical comparison applies SciPy’s sparse trust-region interior-point method to the unfolded convex quadratic problem (Virtanen et al., 2020). The primal promise method performs backward dynamic programming directly on the public graph, restricting successor promises to a grid while allowing each current action to be the continuous residual. None of these baselines receives the quotient optimum as an input.

Each case–method pair runs in a fresh sequential worker. Three repetitions are retained with their minimum, median, and maximum. Numerical-library imports are outside the timed method calls and are performed in all workers. Peak memory is an absolute process high-water mark, including the common runtime and libraries; it does not estimate allocator-level incremental memory. The comparison suite repeats construction, root inversion, certificate generation, and the inherited whole-response audit separately. In the scale suite, construction, inversion, and machine minimization are repeated three times; certificate generation and an independent *single-query* policy audit are each timed once and explicitly identified as diagnostics. The latter checks conditional caps, exact state flow, stationarity, complementarity, root payment, and reward without checking every affine response segment. The companion specifies all tolerances, generators, and measurement boundaries.

6.2 Exact tree agreement and a public-graph primal alternative

Table 1 reports the unfoldable comparisons. All eight tree optima equal the quotient optima as exact rational numbers. The numerical convex solves terminate successfully; their largest absolute value difference is 3.64×10^{-7} and their largest primal constraint residual is below 2.63×10^{-14} . A small residual is not an exact optimality certificate: the rational quotient/tree comparison and the separate certificate audits provide a different type of evidence.

For these implementations and inputs, the tree-to-quotient timing ratio ranges from approximately 1.9 to 40.7. These are comparisons with a specified occurrence-indexed baseline, not a performance bound against the best laminar solver. A normalized memoized version of the tree reduction has the same recurrence as the quotient compiler; no speed advantage over that equivalent implementation is claimed. The generic convex comparison also concerns an explicitly unfolded formulation, not every possible compact reformulation.

Table 2 provides the more important structural control: a competing algorithm that already exploits the public graph. Its policies satisfy the exact root promise and all continuation caps under rational replay. At resolution 80, the largest absolute loss among the six reported horizon–seed pairs is below 9.77×10^{-5} . One instance is already exact at resolution 40; others retain a positive loss. The gain from the dual representation in this study is exact continuous-response closure and an exactly certified policy without selecting a promise mesh, rather than simply avoiding history enumeration.

We do not label the promise-grid implementation as SDDP. An SDDP performance experiment would additionally specify initialization, sampling, cut selection, stopping rules, and comparable error guarantees. Counting a hypothetical scenario tree does not supply that experiment. Our public-graph grid provides a concrete state-method baseline for the stated scalar problem; results here do not rank stochastic dual decomposition implementations on larger, less structured convex models.

6.3 Graph scale, response density, and rational precision

Tables 3 and 4 separate three different sources of cost. The broadly capped coupled family has heterogeneous payment and reward coefficients and rational transitions. Its maximum coefficient length rises from 198 to 388 to 768 bits as the horizon doubles from 16 to 32 to 64 dates. A tighter-cap 128-date instance reaches

1,501 bits. This tested growth is approximately linear in horizon; it is not asserted to establish an asymptotic growth law for arbitrary inputs.

The late-renewal family retains heterogeneous, binding caps at the penultimate date and recombines afterward. It reaches 2,045 public vertices and 8,164 edges. It has ten global response knots, 22,419 stored segments, and a largest coefficient length of 2,213 bits. The optimal protocol requires four writable symbols. Construction takes a median 2.62 seconds, while the separate certificate-generation and single-query audit measurements take 0.87 and 1.19 seconds. This is a long rational-horizon example with sparse cap activation; it is not presented as a worst-case response-density test.

The complementary dense-knot family has slack continuation caps and distinct local clipping thresholds. At 511 vertices it stores 197,369 affine segments, compared with 3,161 at 63 vertices. It exposes approximately quadratic table growth even though coefficient lengths remain small and execution needs only one symbol. The difference between this case and the late-renewal case makes the distinction between a global knot universe, stored response segments, coefficient precision, and execution memory visible in measured data. We also retain a censored exploratory tight-cap run that did not complete its full repeated protocol within the imposed run limit; it is not treated as a successful large-instance timing.

A compiled root curve supports repeated inversions without rebuilding all response tables. The companion reports 101 evenly spaced root queries for each late-renewal size. Those times include inversion only; emitting and certifying a new policy for each promise is an additional operation. The read-only response tables and their numerical precision remain charged separately from the minimized writable alphabet.

6.4 Verification and the retained evidence boundary

All 35 inherited exact cases reproduce their complete response curves and values, with 147,157 assertion groups in the inherited independent verifier. Forty additional small graph instances agree exactly with the independently implemented tree reduction. Two hundred and two comparisons of explicitly expanded continuation plans check the behavioral equivalence; a separate case reduces two distinct incoming prices to a single necessary symbol. Exhaustive enumeration checks the ordered frontier against 2,308 arbitrary set partitions, including unequal probabilities and nonquadratic costs. Boundary tests cover 150 feasible corridor anchors, 96 tied-normal occurrences, and 1,185 exact supporting inequalities. The test suite also rejects deliberately invalid inputs and altered certificates. These are implementation and finite-instance checks, not independent mathematical proofs of universal claims.

The earlier deployment study is neither removed nor used as favorable algorithmic evidence. At the strict tolerance, 83 of 84 cache queries refreshed and no cache speed advantage was measured. Switching paths, comparator-adjusted rents, continuous-state envelopes, correlated certificates, and governance results retain their original statements, assumptions, proofs, and numerical records in the supplementary theory volumes. The new price compiler and the unfavorable cache experiment test different algorithms and different questions.

7 Conclusions

Classical laminar allocation and a compact public-graph implementation are two representations of the same accepted service problem. The quotient theorem identifies exactly when their normalized responses coincide; the coefficient-event argument supplies graph-input and rational bit bounds. The backward behavioral quotient answers a different operational question: which reached prices genuinely require distinct stored symbols. Cap heterogeneity can make the physical public state insufficient even when a much smaller price record suffices.

For renewal agreements, the value of a limited alphabet admits an ordered allocation formula under heterogeneous probabilities and costs. Increasing outside-option dispersion at a fixed mean payment raises that value, while the exact alphabet requirement can jump at arbitrarily small dispersion. This separates the engineering requirement for exact execution from the economic value of approximate retention.

These conclusions rely on a scalar additive commitment and separable rewards. They do not assign the same finite-price guarantee to switching, coupled commitments, or continuous outer shared-table design. Those models and the previously established adaptation, sensitivity, and deployment results retain their own assumptions and proofs in the supplementary theory volumes. The computational evidence establishes exact agreement on tested instances and comparisons for explicitly implemented methods; it does not establish universal algorithmic dominance or empirical service calibration.

Code, data, and reproducibility. The accompanying review package includes the source modules, exact input/certificate files, independent auditors, raw repeated timings, and a README under `revisions/or-r30-quotient-memory-20260923`. The electronic companion specifies every benchmark and audit scope. The retained R29 manuscript and companion, earlier derivations, and adverse cache evidence are supplied without deletion. The reference implementation uses Python exact rational arithmetic; SciPy is needed only for the numerical comparison. No external dataset is required for these synthetic experiments.

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Table 1: Comparison on unfoldable instances

Dates	Seed	Vertices	Occurrences	Quotient	Tree	Convex	Value error
4	13	7	15	0.0009	0.0018	0.0265	3.64e-07
4	29	7	15	0.0009	0.0022	0.0697	9.12e-10
6	13	11	63	0.0023	0.0099	0.0713	5.88e-09
6	29	11	63	0.0022	0.0090	0.1183	8.74e-10
8	13	15	255	0.0040	0.0379	0.2627	1.36e-09
8	29	15	255	0.0039	0.0483	0.3436	2.57e-09
10	13	19	1023	0.0041	0.1663	1.2747	1.39e-08
10	29	19	1023	0.0059	0.1670	2.4538	3.22e-10

Times are seconds. Quotient time is the sum of the separately measured phase medians for construction, root inversion, and certificate generation; the exact audit is excluded. Tree and convex times are end-to-end medians of three repetitions. Tree and quotient values agree as rational numbers. Value error is the absolute difference between the numerical convex-solver value and the exact optimum; its feasibility and stationarity residuals are retained separately. All methods use the same input, not an optimum supplied by another method.

Table 2: Primal promise discretization on the public graph

Dates	Seed	$Q = 10$	$Q = 20$	$Q = 40$	$Q = 80$	Seconds	States
4	13	8.82e-03	5.11e-03	0	0	0.0085	619
4	29	3.21e-02	7.48e-03	1.50e-04	5.29e-06	0.0084	599
8	13	2.54e-02	4.00e-03	3.79e-04	7.69e-05	0.0342	2275
8	29	2.19e-02	8.73e-04	2.25e-04	5.95e-05	0.0333	2279
12	13	1.68e-02	1.87e-03	4.15e-04	9.76e-05	0.0792	4971
12	29	2.15e-02	9.97e-04	3.15e-04	5.68e-05	0.0853	4967

The four resolution columns report the exact feasible-policy value loss relative to the quotient optimum. Child promises are multiples of $1/Q$; local actions are continuous residuals and the root promise is not rounded. Seconds and stored states refer to $Q = 80$. Times are medians of three repetitions. The grid is a public-graph dynamic program, not a scenario-tree baseline; exact rational replay checks its feasibility. A zero gap on a particular instance is not a uniform exactness claim for the grid.

Table 3: Response representation and rational coefficient growth

Family	Vertices	Edges	Knots	Segments	Largest	Bits
Coupled	61	228	128	2428	83	198
Coupled	125	484	240	10168	159	388
Coupled	253	996	436	34748	279	768
Tight caps	509	2020	723	67212	295	1501
Late renewal	253	996	10	2706	11	277
Late renewal	509	2020	10	5523	11	554
Late renewal	1021	4068	10	11154	11	1107
Late renewal	2045	8164	10	22419	11	2213
Dense knots	63	122	96	3161	68	12
Dense knots	127	250	192	12473	132	14
Dense knots	255	506	384	49529	260	16
Dense knots	511	1018	768	197369	516	18

Knots counts the distinct global union; segments counts the entire stored family. Bits is the largest numerator or denominator bit length among reduced affine coefficients. Late-renewal caps are heterogeneous and active only at the penultimate date; this family is deliberately distinct from the dense-knot stress family and from the broadly capped coupled family. It tests long rational horizons with a small knot universe, not worst-case quadratic storage.

Table 4: Separate construction, execution, and audit costs

Family	Vertices	Build	Invert	Certificate	Audit	Symbols	Memory
Coupled	61	0.1220	16.3	0.0484	0.0775	18	97.3
Coupled	125	0.5158	183.1	0.1154	0.2222	23	117.5
Coupled	253	2.3536	433.9	0.6680	1.1607	61	227.2
Tight caps	509	7.1123	1426.9	0.5700	0.1233	1	396.5
Late renewal	253	0.0976	19.1	0.0154	0.0293	4	97.2
Late renewal	509	0.2551	24.8	0.0483	0.0839	4	107.4
Late renewal	1021	0.7465	48.9	0.1691	0.2958	4	137.3
Late renewal	2045	2.6156	127.1	0.8716	1.1871	4	251.0
Dense knots	63	0.0417	69.7	0.0043	0.0037	1	94.3
Dense knots	127	0.1679	146.2	0.0163	0.0092	1	100.2
Dense knots	255	0.6180	271.4	0.1215	0.0153	1	121.4
Dense knots	511	2.5074	504.5	0.3282	0.0315	1	214.3

Build, certificate, and audit are seconds; inversion is microseconds. Build and inversion use three repetitions. Certificate generation and the independent single-query policy audit are each measured once in this scale suite; they are diagnostic measurements, not repeated-time claims. Symbols is the exact minimized writable alphabet. Memory is the absolute fresh-worker process high-water mark in MiB, including common Python, NumPy, and SciPy imports; it is not incremental algorithm allocation. Whole-curve audits are reported separately from these single-query KKT/flow audits.